Cattle Biometric System for Identification and Verification

A Master's Project report submitted in partial fulfilment of the degree of five years Integrated Dual Degree program

IDD Computer Science and Engineering

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DECLARATION

I, Tejas Bhausaheb More, a student of Fifth year Integrated Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering at Indian Institute of Technology (BHU), Varanasi, Uttar Pradesh hereby declare that this Master's Project report is submitted by me in partial fulfilment of the requirement of Five year Integrated Dual Degree.

The following report displays my work undertaken as a part of Master's Project during the Ninth semester.

The results embodied in this report have not been submitted to any other Institution/University for the award of this degree.

The information and report submitted is the outcome of my efforts undergone during that period and is original to the best of my knowledge.

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PROBLEM STATEMENT

Individual cattle identification is critical for modern precision livestock farming, enabling automated health monitoring, feeding management, and breeding record maintenance. Traditional methods such as RFID tags are prone to loss or damage, while manual visual identification is time-consuming and error-prone at scale.

This project aims to develop an open-set biometric identification and verification system for cattle using deep learning on multimodal image data (face and muzzle). The key challenge is distinguishing between enrolled cows (known to the system) and unknown cows (never seen before), a critical requirement for real-world deployment where new animals may enter a facility.

Primary objective: To achieve minimal false accept rates to prevent unknown cows from being incorrectly identified as enrolled animals, while maintaining high recognition accuracy for known cows.

METHODOLOGY

Dataset preparation and partitioning -

The dataset contains images of cows. Two biometric modalities were captured for each animal:

- Face images: Frontal facial region of the cattle
- Muzzle images: Close-up images of the nasal pattern

Each individual cow was assigned a unique identifier (e.g., HF-6-C8020, HF-7-C8018).

The dataset was partitioned into three mutually exclusive sets based on cattle identity (not image samples). This is to ensure that no cattle individual appears in multiple splits, simulating arealistic deployment scenario where the system must handle completely unknown animals.

Set	Purpose	Cattle count	Face images	Muzzle images	Total samples
Training	Enrollment	345	3510	3534	7044
Validation	Threshold	30	291	297	588
Test	Final blind evaluation	93	925	925	1850
Total	-	468	4726	4756	9482

Training set (345 cattle): Represents enrolled cattle in the authentication gallery. These are the "authorized" individuals the system must recognize.
Validation set (30 cattle): Completely unknown cattle used for threshold calibration.
Test set (93 cattle): Completely unknown cattle for final blind evaluation. This set is never used during model selection or threshold optimization.

Preprocessing Pipeline (Final Cow Workflow):

All images underwent the following preprocessing and region-extraction pipeline:

- Input format:Raw RGB cattle images collected in real-world farm conditions
- Region of Interest (ROI) Localization (YOLO):
  - AYOLO-based detector isused to obtain the bounding box around the biometric region
  - The system runs ROI extraction separately for:
    - Face region
    - Muzzle region
- Foreground Extraction (GrabCut):
  - GrabCut is applied using the YOLO bounding box as an initialization prior
  - This step suppresses background noise (soil, rope, vegetation, walls, human objects)
  - Output: cleaned foreground crop focused only on cattle biometric textures
- Resizing:
  - Final crops resized to 224×224 using bilinear interpolation
- Color space:
  - RGB (3 channels) maintained
- Normalization:
  - Pixel values scaled to [0,1]
  - Channel-wise normalization applied using ImageNet statistics:
    - mean = [0.485, 0.456, 0.406]
    - std = [0.229, 0.224, 0.225]

This pipeline ensures the network receives consistent inputs while remaining robust to real-world variations.

MODEL ARCHITECTURE (Final Cow Embedding Model)
InputCrop (Face / Muzzle)224×224×3
↓
ResNet-50 Backbone (Feature Extraction)
↓
Feature Map Tensor (deep spatial features)
↓
Projection + Tokenization
↓
Transformer Head (Contextual Feature Refinement)
↓
Global Pooling
↓
Fully Connected Projection Layer → 512-d embedding
↓
L2 Normalization (unit-length embedding)
↓

512-dim Normalized Embedding
↓
[TRAINING ONLY] ArcFace Loss → classification logits (345 identities)

Component Details

1. ROI Extraction: YOLO + GrabCut

YOLO Detector

- Purpose: Localize cattle biometric region under varying background conditions
- Output: bounding box coordinates for face/muzzle ROI
- Benefit: reduces search area and removes irrelevant visual content

GrabCut-Segmentation

- Purpose: refine YOLO crop by removing non-biometric pixels
- Initialization: bounding box from YOLO used as GrabCut rectangle
- Benefit: improves embedding quality by forcing the model to learn cattle texture patterns rather than background patterns

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2. Feature Extraction Backbone: ResNet-50

- Architecture: ResNet-50
- Strength: residual blocks preserve gradient flow and enable deep feature learning
- Output: high-level cattle identity features (robust texture + shape representation)

ResNet-50 forms the primary feature generator for both face and muzzle pipelines.

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3. Transformer Head (Contextual Refinement)

- Purpose: refine CNN feature maps into stronger identity-specific representations
- Mechanism: self-attention learns spatial relationships within the biometric region
- Benefit: improves discrimination under:
  - pose changes
  - partial occlusion
  - low-quality crops
  - intra-class variation

The transformer head upgrades raw CNN features into context-aware embeddings.

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4. Dense Projection Layer (Embedding Head)

- Output embedding size: 512dimensions
- Purpose: compact identity vector representation for matching and retrieval
- Enables:
  - fast similarity computation
  - scalable gallery search across enrolled cattle

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5. L2 Normalization

- Operation:

  $embedding_normalized = embedding / ||embedding||_2$
- Forces embeddings onto a unit hypersphere

Purpose:

- standardizes embedding magnitude
- supports cosine similarity based retrieval
- stabilizes threshold decisions

Training Objective (Arc Face Loss)

During training, the system learns discriminative embeddings using Arc Face supervision.

Arc Face Loss

- Input: normalized 512-d embeddings
- Output: classification logits over 345enrolled cattle identities
- Key effect: introduces angular margin between identities

Arc Face enforces:

- tighter intra-class clustering (same cow embeddings stay close)
- stronger inter-class separation (different cows become angularly distant)

Gallery Formation (Face Gallery + Muzzle Gallery)

The final system is gallery-driven.

Two galleries are constructed:

1. Face Gallery
    - Stores one template embedding per enrolled cow identity
    - Structure: { cow_id : face_template_embedding }
2. Muzzle Gallery
    - Stores one template embedding per enrolled cow identity
    - Structure: { cow_id : muzzle_template_embedding }

Templates are formed by aggregating multiple samples for each cow (mean embedding) and storing the final normalized vector.

Similarity Metric(Cosine Similarity)

Matching is done using cosine similarity:

similarity(e1, e2) = e1 · e2

Because embeddings are L2-normalized, cosine similarity reduces to a dot product which is:

- fast to compute
- consistent across samples
- directly aligned with Arc Face training geometry

Open-Set Decision Using Thresholds (Fusion-Based)

The system is designed for an open-set deployment scenario, where the input cattle may either belong to the enrolled gallery (known cattle) or may be completely unseen (unknown cattle)

To handle this, the system does not rely only on classification logits. Instead, it applies similarity thresholding in embedding space.

Similarity Thresholds Used

The inference pipeline uses three thresholds:

- t_face : threshold for face similarity score
- t_muzzle : threshold for muzzle similarity score
- t_fusion : threshold for final fusion score

These thresholds control when a cattle sample is accepted as a valid known identity and when it is rejected as unknown.

SIMILARITY COMPUTATION (Face and Muzzle)

For both face and muzzle images:

1. The input crop is converted into a 512-dimensional L2-normalized embedding
2. The embedding is matched against the corresponding gallery embeddings
3. The highest similarity identity is selected as the best candidate

Cosine similarity is computed as:

$$S = \boldsymbol{etest}.\ \boldsymbol{egallery}$$

Where:

- $\boldsymbol{etest}$ is the normalized embedding of the test sample
- $\boldsymbol{egallery}$ is the normalized embedding template from the gallery

The similarity score lies in the range $[-\mathbf{1},\mathbf{1}]$, where higher values indicate stronger identity match confidence.

FINAL INFERENCE PIPELINE: WEIGHTED FUSION VERIFICATION

Unlike a sequential two-factor gate, the final deployed workflow uses weighted fusion of face and muzzle confidence scores.

Input:

Face Image + Muzzle Image

↓

Face Feature Extraction → Matching (1:N)

↓

Muzzle Feature Extraction → Matching (1:N)

↓

Score Fusion (Weighted Combination)

↓

Final Threshold Decision (Open-Set Accept / Reject)

Step 1: Face Recognition (1:N Matching)

- A face embedding is extracted using the face embedding model
- The face embedding is compared against all enrolled face templates
- Best candidate identity is selected:

$$Sface = \max\ (\text{cosine}(eface, Gface))$$

If the face similarity is below a threshold:

- If $Sface < tface \rightarrow$ the face evidence is weak

Step 2: Muzzle Recognition (1:N Matching)

- A muzzle embedding is extracted using the muzzle embedding model
- The muzzle embedding is compared against all enrolled muzzle templates
- Best candidate identity is selected:

$$S_{muzzle} = \max\ (\text{cosine}(e_{muzzle}, G_{muzzle}))$$

If the muzzle similarity is below a threshold:

- If $S_{muzzle} < t_{muzzle} \rightarrow$ the muzzle evidence is weak

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Step 3: Fusion Score Computation (60:40)

The final decision is made by combining both biometric score.

$$S_fusion = w_face \cdot S_face + w_muzzle \cdot S_muzzle$$

Where:

- $w_{face} = 0.6$
- $w_{muzzle} = 0.4$

This ratio is chosen to prioritize face similarity while still using muzzle similarity as a stabilizing biometric signal.

Step 4: Open-Set Accept / Reject Decision

The final open-set decision is taken using the fusion threshold:

- If $S_{fusion} \geq t_{fusion} \rightarrow$ ACCEPT
- Else $\rightarrow$ REJECT

This ensures that even if one modality is moderately strong, the overall decision depends on combined evidence and is resistant to false accepts.

Why Weighted Fusion Improves Open-Set Robustness

Weighted fusion improves system robustness because:

- Face similarity alone can be unstable in cases of:
  - Pose changes
  - Partial occlusion
  - Illumination noise
- Muzzle similarity provides additional biometric consistency
- Combining both reduces the probability of falsely accepting unknown cattle

In practical deployment, the fusion strategy ensures that high confidence is required across the biometric modalities rather than relying on a single cue.

OUTPUT FORMAT (Inference Results)

For each test sample, the system outputs:

- True identity (known only for evaluation sets)
- Predicted identity (if accepted)
- Face similarity score

- Muzzle similarity score
- Fusion similarity score
- Final decision (ACCEPT / REJECT)

This output supports evaluation metrics such as:

- False Accept Rate (FAR) → unknown accepted as known
- True Accept Rate (TAR) → known correctly accepted

TRAINING:

- The Final Cow system trains embedding-based biometric models for cattle identification.
- Instead of outputting final class labels during inference, both models generate 512-dimensional normalized embeddings, which are later used for similarity matching against an enrollment gallery.
- Two independent embedding models were trained:
  - Face Embedding Model
  - Muzzle Embedding Model
- Both models follow the same embedding-learning strategy but are trained on their respective modality-specific crops.

Training Objective: ArcFace-based Metric Learning

Training is performed using ArcFace Loss, which enforces separation in angular space.

- Goal: produce well-separated identity clusters for enrolled cattle
- Embedding space: 512-dimensional unit hypersphere
- Effect: same-cow embeddings move closer, different cows move farther apart

ArcFace strengthens cluster margins, which improves the reliability of cosine similarity matching in deployment.

Model Output Used in Inference:

During inference,the classification layer is only this is used:

- EmbeddingModel(image) → 512-d normalized vector

This embedding is compared to the gallery using cosine similarity.

Models Stored After Training

In the final implementation, both trained models were stored as .pt checkpoints and loaded using:

```
ckpt = torch.load(model_path)
state_dict = ckpt["model_state"]
model.load_state_dict(state_dict)
```

This is important because the checkpoint is not a raw Torch-Script model, it is a dictionary with model weights stored under the key model_state.

Backbone Architecture Used

The deployed embedding network uses a ResNet-50 backbone:

- ResNet-50 feature extractor outputs a pooled 2048-d feature vector
- A dense layer projects it into a 512-d embedding space
- Embeddings are L2-normalized

This is the exact inference structure used:

Block	Output
ResNet-50 CNN backbone	-> 2048 features
FC Projection Layer	->512 embedding
L2 normalization	->Unit-length embedding

Enrollment Gallery Construction

Instead of storing embeddings in arrays, the final cow gallery was saved as a dictionary structure inside .npy.

Actual gallery file content format

The .npy file stores a single object:

- shape = ()
- dtype = object
- content = dict

Example Structure:

```
{
  "DEONI-19-C2082": np.array([512 floats]),
  "GB-5-C8012": np.array([512 floats]),
  ...
}
```

Two galleries are stored separately:

- face_gallery.npy
- muzzle_gallery.npy

Gallery Loading Logic Used

The evaluation loads them using:

```
obj = np.load(path, allow_pickle=True).item()
```

Then both galleries are aligned using common cow IDs:

- intersection of keys from face and muzzle dict
- sorting ensures deterministic ordering

The final gallery matrices are built as:

- ids → list of cow IDs
- face_gallery_matrix → (N,512)
- muzzle_gallery_matrix → (N,512)

Gallery Type	Total Identities
Face Gallery IDs	345
Muzzle Gallery IDs	345
Common IDs Used for Evaluation	345

Example enrolled identities:

- DEONI-19-C2082
- GB-5-C8012
- GR-6-C2004
- GRI-4-C25
- GRI-5-C12

TEST SET GENERATION (Image-Level Evaluation):

The test dataset used in this section is not a single-image-per-cow dataset.
Instead, the evaluation is done on image-level pairs:

- Each test sample = one face image + one muzzle image

Test Folder Structure

Unlike flat test folders, the test set is organized as folder-of-folders:

```
FACE_TEST_ROOT/
  cow_id_1/
    img_1.jpg
    img_2.jpg
    ...
  cow_id_2/
    img_1.jpg
    ...

MUZZLE_TEST_ROOT/
  cow_id_1/
    img_1.jpg
    img_2.jpg
    ...
```

The test loader dynamically builds pairs by:

1. Iterating each cow folder
2. Listing all images inside face and muzzle directories
3. Pairing images using index-based matching (pairing="index")

Test Pair Statistics

Total cows in test set	Total paired test samples
93	916

This indicates that the evaluation is performed on 916 independent authentication attempts, providing a stronger estimate of open-set behavior.

Inference Workflow Used In Results Section

The test-time workflow is based on cosine similarity matching against the enrolled gallery.

For each test pair:

Step 1: Face Similarity Score

- compute face embedding

- compute similarity against all face templates (1:N)
- take max similarity:

$$S_{face} = \max(e_{face} \cdot G_{face})$$

Step 2: Muzzle Similarity Score

- compute muzzle embedding
- verify only the predicted cow's muzzle template (1:1)

  - **S_muzzle = e_muzzle · G_muzzle[predicted_idx]**

OPEN-SET EVALUATION (Unknown-only)

The test set used here is open-set, containing only unseen cattle identities.
Therefore:

- there are no "known positives" in test
- TAR is not a meaningful metric here
- the most critical metric becomes FAR

Metrics Computed

False Accept Rate (FAR)
Where:

- FP = number of unknown cattle samples accepted as known
- Total Unknown Attempts = 916

True Accept Rate (TAR)
Since there are no genuine enrolled cows in this test set:

- TP = 0
- TAR = 0 by definition for unknown-only evaluation

RESULTS

The evaluation outputs only:

- TP
- FP
- TAR
- FAR

Because the test set contains only unknown cattle.

Metric	Value
Total Attempts	916
TP	0
FP	0
TAR	0.0000
FAR	0.0000

Interpretation Since FP = 0, the system successfully rejects all unknown cattle and demonstrates strong robustness.

Open-set

The output JSON confirms that rejected samples are returned with:

- pred = None
- "reason": "Face gate failed" or "reason": "Muzzle gate failed"

This indicates that the gallery matching + similarity thresholding is working as intended for unseen identities.

New Open-Set Evaluation on Structured Test Dataset

Test Set Preparation

For this evaluation, a new open-set test dataset was constructed and organized explicitly for identity-level verification.
The dataset was orchestrated and structured to ensure consistency between modalities.
Each cattle identity was stored under a single folder, with two subfolders:

- face containing all face images of that cattle
- muzzle containing all muzzle images of the same cattle

This structure ensured that all face and muzzle samples belonging to a cattle identity were grouped together, allowing both image-wise and cow-wise evaluation without ambiguity.
Importantly, none of the cattle identities present in this dataset were included in the training gallery, making this a strictly unknown-only open-set test.

Gallery and Model Usage

The face and muzzle galleries were constructed exclusively from the training set.
These galleries represent the enrolled known identities.
The face and muzzle embedding models were trained using the same training data, after which the trained weights were saved and frozen.
During this evaluation:

- the saved models were loaded without modification
- the training galleries were reused as-is
- the new dataset was used only for testing

This setup ensures a clean separation between training and testing.

Evaluation Protocol

Since the test set contains only unknown cattle identities, the evaluation follows an open-set unknown-only protocol.
Decision flow:

1. Face embeddings are compared against the training face gallery (1:N)
2. If the face similarity score passes the face threshold, muzzle verification is performed (1:1)
3. A sample is accepted only if both face and muzzle thresholds are satisfied

Because there are no genuine known identities in this test:

- True Accept Rate (TAR) is not meaningful
- False Accept Rate (FAR) is the primary metric

Metrics Computed

- False Positives (FP)
- False Accept Rate (FAR)
- By definition, TP = 0 for unknown-only evaluation
- True Accept Rate (TAR)

Results Summary (New Open-Set Test)

FAR = 0, FP = 0.

Interpretation

Since FP = 0, the system successfully rejected all unknown cattle identities in the new test set.

This indicates that:

- the similarity thresholds are well-calibrated
- the gallery-based matching does not falsely associate unseen identities
- the face-muzzle gated verification pipeline behaves correctly under strict open-set conditions

This confirms that the system's rejection behavior is deliberate and threshold-driven, rather than accidental.